P1-22 — Adoption Recommendation for the Proposed Consequences (Integration-Readiness Check)

Inputs: P1_20_NEW_CONSEQUENCES.md (results R1–R4), P1_21_PROOF_AUDIT.md (independent re-derivation; all four correct with scope fences), the manuscript P1_15_MANUSCRIPT_DRAFT.md/.tex, the frozen blueprint (02_THEOREM_BLUEPRINT.md, SHA-256 ec2fd8d58f8d2a986314f83997d750d689f7c88e98ec8f87753ab4c9cf438cff, unchanged), P1_15_OPTIONAL_COROLLARIES.md (N1–N3), and the gap register. Repository at start: commit dffd829, local = remote, clean.

This file is a recommendation only. It is not an approval, and it edits neither the frozen blueprint nor the manuscript. Every candidate remains PROPOSED until a dated Commander decision (GAP-17 rule). No novelty claim is made; **Gate G1 remains DEFERRED** (L1 unread at theorem level, P1-19), so nothing here may be worded as "new" or "first" in the manuscript sense — the classifications below concern only mathematical correctness, non-duplication, and fit.

Notation plan for any future integration (binding on the plan, to remove the P1-21 overload flag):

- Delta := n ∗ sum_s m_s — the maximum/ambient F_q-dimension (never bare D, which is reserved for the dual code D(C)).
- N_SO(K) — number of self-orthogonal labeled assignments with dim C = K (equivalently joint type (K,K)).
- N_DC(K,H) — number of dual-containing labeled assignments with (dim C, dim Hull) = (K,H).
- a_O := P ∗ w_O ∗ L_O (the per-orbit LCD/Bernoulli weight; note sum_O a_O = Delta).
- s_L^{(P,w)}(u,z) := tr((T^{(P)}_w(u,z))^L) if R4 is ever displayed.

Standing assumptions for all statements below: the manuscript's hypotheses — A = prod_s K_s square-free affine with labeled components, common n = P n_0, gcd(n_0,p)=1, compatible twists lambda_s^{1+p^{d_s-k}}=1, code-first pairing, labeled exponent assignments; probabilistic statements use the uniform labeled model. Per orbit: K(alpha) = w sum_i (P-alpha_i), H(alpha) = w sum_i min(P-alpha_i, alpha_{i-1}), indices mod L.

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1. R1 — Joint complement–reversal symmetry → RECOMMENDED for adoption

Exact statement (manuscript-ready form). Let Delta = n ∗ sum_s m_s. For all integers K,H, #{a : (dim C(a), dim Hull(a)) = (K,H)} = #{a : (dim C(a), dim Hull(a)) = (Delta-K, H)}, equivalently E(u,z) = u^{Delta} E(u^{-1}, z). Consequently, whenever Pr(dim C = K) > 0, the conditional law of dim Hull given dim C = K equals that given dim C = Delta-K.

Verification of Delta against the manuscript's notation. Per component s, sum_{O in O_s} w_O L_O = m_s ∗ sum_{f in F_s} deg_{K_s} f = m_s n_0, because F_s is the set of distinct monic irreducible factors of the degree-n_0 polynomial x^{n_0} - mu_s (manuscript Section 2). Hence sum_O P w_O L_O = P n_0 sum_s m_s = n sum_s m_s = Delta. This matches the manuscript exactly: dim_{F_q} sums carry the factor m_s per component (Theorem 5.2), each component code has K_s-dimension at most n, and dim C + dim D(C) = Delta — so Delta is both the maximum code dimension and the ambient dimension, as the symmetry requires. The map Phi(alpha)_i = P - alpha_{-i} is an involution of the full labeled assignment space (orbitwise product), sends dim C to Delta - dim C by the cyclic-sum computation, and preserves dim Hull termwise after the reindexing j = -i+1 (P1-21 Section 1: proof complete under every stated hypothesis; L=1, P=1, w>1 edge cases verified; the polynomial identity needs no field hypothesis beyond T5/T6 themselves).

Proof location: P1-20 Section R1; independently re-derived in P1-21 Section 1 (verdict: correct, complete). **Validation evidence:** code/validate_p1_20.py (grid P in {1..4}, L in {1..4}, w in {1,2}, coefficientwise); P1-21 fresh parameters (P,L,w) = (5,3,3), (5,5,1), (6,4,2), (7,3,3), (4,5,2) (the P=6 case tests the polynomial identity only). All passed; evidence only, proof is the involution.

Relationship to existing results: C2 is the H-summed (marginal) shadow; no frozen corollary states the joint symmetry or the conditional-law consequence. No duplication. **Recommended placement:** Section 6, as a corollary immediately after Theorem 6.3 (integration phase handles the automatic renumbering of Proposition 6.4). One paragraph statement + involution proof; the conditional-law sentence as a second short corollary or remark. **Decision: RECOMMENDED** — strongest of the candidates: complete elementary proof, genuinely joint content, useful (halves independent joint coefficients; explains the symmetric conditional pattern).

2. R2 — Restricted SO/DC enumerators → RECOMMENDED for adoption

Exact statement (with cleaned notation). Per orbit define S_w(t)_{a,b} = t^{w(P-b)} if a+b >= P else 0, and G_w(u,z)_{a,b} = u^{w(P-b)} z^{w a} if a+b <= P else 0 (row a = previous exponent, column b = current, the T5/Theorem 6.1 convention; the closing edge in the trace is the orbit-boundary edge). Then sum_{a self-orthogonal} t^{dim C(a)} = prod_s prod_O tr(S_{w_O}(t)^{L_O}) (joint form at t = u z, since dim Hull = dim C there), and sum_{a dual-containing} u^{dim C(a)} z^{dim Hull(a)} = prod_s prod_O tr(G_{w_O}(u,z)^{L_O}). Moreover N_SO(K) = N_DC(Delta-K, K) for all K.

Verification against C4 and the transfer convention. C4 (Corollary 7.4) gives the edge criteria a+b >= P (SO) / <= P (DC) and, via dim D(C) = sum_O w_O sum_i alpha_i, the hull dimension on DC tuples; the matrices attach exactly the per-edge code weight w(P-b) and, for DC, the dual/hull weight w a from the previous vertex, whose cyclic sum is w sum_i alpha_i. Boundary cases confirmed in P1-21: L=1 reduces to the diagonal conditions 2a >= P / 2a <= P; u=z=1 collapses both sides to the C4 counts prod tr(B_P^{L_O}), prod tr((B'_P)^{L_O}). The P1-20 (P=2,L=2) example was reproduced by hand in P1-21 (SO {(0,0):1,(1,1):2,(2,2):3}; DC {(4,0):1,(3,1):2,(2,2):3}, with N_DC(4-K,K) = N_SO(K)). The bare letter D for counting functions is **removed** from this plan (N_SO, N_DC above; Delta for the constant), resolving the P1-21 notation flag.

Proof location: P1-20 Section R2; re-derived in P1-21 Section 2 (correct; orientation/boundary verified). **Validation evidence:** P1-20 validator (restricted traces = brute force on the grid, including H=K on SO and H = w sum alpha on DC tuples); P1-21 fresh cases (5,3,3), (6,4,2), (7,3,3). All passed. **Relationship:** a distribution-level refinement of C4 (same closed-walk mechanism; C4 is exactly the u=z=1 specialization, to be stated as such). Not a restatement, provided the text presents the ∗distributions∗ as the content and the count collapse as a check. **Recommended placement:** Section 7, as a new corollary immediately after Corollary 7.4, including the N_SO(K) = N_DC(Delta-K,K) symmetry and the P=2,L=2 counts as a worked line. **Decision: RECOMMENDED** — genuine new distributions, complete proof, natural fit next to C4.

3. R3 + N3 — one unified LCD candidate → RECOMMENDED for adoption (as a single result only)

Unified exact statement. (i) (N3 form) sum_{a : Hull(a)=0} u^{dim C(a)} = prod_O (1 + u^{a_O}), a_O = P w_O L_O. (ii) (R3 form) Under the uniform labeled model conditioned on Hull(a) = 0, dim C =d sum_O a_O xi_O with xi_O independent Bernoulli(1/2); hence E[dim C | LCD] = Delta/2 and Var(dim C | LCD) = (P^2/4) sum_O (w_O L_O)^2.

Verification. The LCD criterion (C3/Corollary 7.3: hull zero iff every orbit is constant 0 or P) was re-audited in P1-21 for all L and all P >= 1; conditioning a uniform product distribution on this product event gives per-orbit independence and the two equally likely options with code contributions a_O and 0. The Bernoulli parameters a_O = P w_O L_O are correct for arbitrary P (odd allowed — parity never enters; contrast the parity-split L=1 hull variance in C6, a different quantity), arbitrary L_O, and arbitrary weights w_O >= 1; boundary cases P=1, L=1, and the degenerate zero-orbit case are consistent (P1-21 Sections 3.1–3.2). sum_O a_O = Delta by the same factor-degree computation as in Section 1 above, so the mean Delta/2 is in manuscript notation.

Proof location: P1-20 Section R3 (= P1-15 companion N3); re-derived P1-21 Section 3. **Validation**

evidence: P1-20 validator (E5-type configuration: LCD count 16, mean 21 = Delta/2, variance 185); P1-21 odd case P=5, (L,w)=(1,2),(2,1): count 4, mean 10, variance 50, matching formulas. **Relationship:** C3 is the u=1 count. R3 and N3 are **the same result** in unnormalized vs. normalized form — presenting them as two contributions would be duplication and is **rejected** (Section 5). The unified corollary's added value over C3 is the full conditional law. **Recommended placement:** Section 7, as a single corollary immediately after Corollary 7.3, stated in the order (i) generating function, (ii) conditional law with mean/variance. **Decision: RECOMMENDED as one unified corollary** (low-depth but exact and quotable; the conditional-law form is the useful one).

4. R4 — Orbit-length recurrence → DEFERRED (supporting proposition; not for the main text now)

Exact statement. For fixed P,w and s_L = s_L^{(P,w)}(u,z) (with the formal s_0 := P+1), there exist c_1,...,c_{P+1} in Z[u,z], independent of L, with s_{L+P+1} = sum_j c_j s_{L+P+1-j} for all L >= 0 (characteristic polynomial of the (P+1) x (P+1) matrix T^{(P)}_w, Cayley–Hamilton). At P=1 only: s_{L+2} = (1+u) s_{L+1} - u(1-z) s_L, s_0=2, s_1=1+u, with s_L = lambda_+^L + lambda_-^L, lambda_{+/-} = ((1+u) +/- sqrt((1-u)^2+4uz))/2 (general w by (u,z) -> (u^w,z^w)).

Assessment. Correct (P1-21 Section 4; P=1 independently verified, including s_2 = 1+u^2+2uz by direct squaring and by enumeration). But per task item 4: the general recurrence is the ∗generic∗ Cayley–Hamilton fact for any matrix; P1-20/P1-21 demonstrate no concrete enumerator consequence in the manuscript beyond faster long-orbit computation, and for P >= 2 there is no closed form (coefficients are computable, e.g. via Newton identities, which is not a closed form; minimality of the order is not claimed). The explicit closed form is therefore **restricted to P=1**, the already-transparent simple-root case. **Recommended placement:** none in the main text at this time. Keep as a supporting proposition in the P1-20/P1-21 record; if the integration phase wants it, at most a two-line remark after Theorem 6.1 ("at fixed P the traces are linear-recurrent in L of order at most P+1; at P=1, ...") with the P>=2 closed-form wording prohibited. **Decision: DEFERRED** — correct but generic; revisit only if a concrete application (e.g. a long-orbit family where the recurrence is actually used) is added.

5. Rejected items and counterexample record (do not integrate in any form)

1. **Presenting R3 and N3 as two results** — rejected: duplication (Section 3).
2. **"Conditional hull mean at fixed code dimension is constant"** — false: P=1,L=3,w=1, joint distribution {(0,0):1,(1,1):3,(2,1):3,(3,0):1} gives conditional means 0,1,1,0 despite Cov=0 (N1). The pattern is exactly the R1 symmetry — a further reason R1 is the right way to say something here.
3. **Independence of dim C and dim Hull** — false: same distribution, Pr(K=2)Pr(H=1) = 9/32 != 3/8 = Pr(K=2,H=1). N1 wording must remain "uncorrelated".
4. **Palindromy in z** — false: hull marginal {0:2, 1:6} in the same example.
5. **A general (P >= 2) closed form for tr(T^L), or minimality of the R4 order** — not established; prohibited wording (Section 4).
6. **Bare D for the maximum dimension or DC counting function** — rejected notation (dual-code clash); Delta, N_DC mandated in Section 1's notation plan.
7. Naive coefficient monotonicity in L and Perron–Frobenius growth **as results** — ill-posed resp. standard background (P1-20 Section 7, P1-21 Section 7); not integrated.

6. Summary of recommendations

Candidate | Decision | One-line reason

R1 (joint symmetry + conditional law) | **Recommended** | Complete involution proof; genuinely joint content beyond C2; clean placement after Theorem 6.3

R2 (SO/DC restricted enumerators + symmetry) | **Recommended** | Complete proof; full distributions refining C4's

counts; fits after Corollary 7.4; notation cleaned (N_SO/N_DC, Delta)
R3/N3 unified (LCD conditional law) | **Recommended (single corollary)** | Exact Bernoulli law/mean/variance from C3; useful form of N3; duplication avoided by unifying
R4 (orbit-length recurrence) | **Deferred** | Correct but generic Cayley–Hamilton; explicit closed form P=1 only; no demonstrated concrete consequence yet
Items in Section 5 | **Rejected** | False (with counterexamples), duplicative, or prohibited wording/notation

Also recorded for completeness: N1/N2 (covariance package) were **not** re-decided in this phase — they remain candidates in P1_15_OPTIONAL_COROLLARIES.md with the P1-20 Section 6 assessment (clean but modest; "uncorrelated, not independent" fencing mandatory). Validators relevant to this recommendation were re-run for this phase: validate_p1_20.py (exit 0, output byte-identical to its saved record) and validate_p1_14.py (exit 0, byte-identical), confirming the frozen core and the candidate evidence base are unchanged. If the Commander adopts any item, integration must proceed by a dated blueprint correction note under GAP-17 with the notation plan above; until then the blueprint, the manuscript, and all statuses are exactly as they were.